

collaborate with a simulated Naïve UCB agent for stacking 20 burgers to proceed forward in the exploration stage so that the robot’s algorithm will be more critical for performance. After these 20 burgers, the simulated agent is replaced with the study participant for preparing 20 more burgers with each algorithm.

The user interface aided the participants by providing information about: the number of satisfied and unsatisfied customers for each burger configuration, the total number of burgers stacked, the configuration of the latest burger and whether it made the customer satisfied.

For a fair comparison, we randomized the reward means only between the users and not between the algorithms. We swapped the actions to prevent participants from realizing they are dealing with the same problem instance. Hence, between the two sets, for example, $\mu_{(0,1)}$ of the first set was equal to $\mu_{(1,0)}$ in the second. To further avoid any bias due to ordering, half of the participants first worked with Naïve UCB and the other half with the Partner-Aware UCB.

Dependent Measures. We measured cumulative regret and the total number of satisfied customers. We excluded the first 20 simulated time steps for a fair comparison. Additionally, the participants took a 5-point rating scale survey (1-Strongly Disagree, 5-Strongly Agree) consisting of 5 questions for each algorithm: “I was usually able to stack the burger I wanted” (*Ability*), “The robot insisted on some sub-optimal burgers” (*Insisting*), “The robot was easy to collaborate with” (*Easy*), “The robot was annoying” (*Annoying*), and “I could get more happy customers if I were stacking burgers alone” (*Alone*).

Hypotheses.

H1. *Users interacting with Partner-Aware UCB robot will incur smaller regret and keep the customers more satisfied.*

H2. *Users will subjectively perceive the Partner-Aware UCB robot as a better partner who can effectively collaborate with them.*

Results-Objective. Partner-Aware UCB achieves lower regret (2.7 ± 0.31) compared to Naïve UCB (3.6 ± 0.31) with statistical significance ($p < .005$). Fig. 8 (left) shows the cumulative regret incurred over time with both algorithms.

The significant difference in the cumulative regret is also reflected in the number of satisfied customers supporting **H1**: Partner-Aware UCB achieved significantly higher number of satisfied customers (13.5 ± 0.6) than Naïve UCB (12.7 ± 0.6), with $p < .05$.

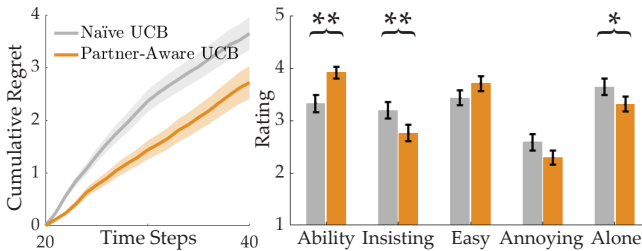


Figure 8: (Left) Average regret over time, (right) survey results for the burger-stacking robot experiment. Single and double asterisks indicate $p < .05$ and $p < .005$, respectively.

Results-Subjective. The Naïve UCB robot often decides stacking an under-explored burger and insists on the same action until that burger is made. While this occasionally helps the humans to exploit a burger, it often causes deadlock situations where both agents are unable to stack their intended burgers. On the other hand, Partner-Aware UCB keeps a model of the human, and avoids such situations. We believe this explains the subjective preferences of the users.

We plot the users’ survey responses in Fig. 8 (right). The responses were reliable with Cronbach’s alpha > 0.95 . The users indicated they were able to stack the burger they wanted (*Ability*) more frequently with the Partner-Aware UCB ($p < .005$), and thought it was easier to collaborate with (*Easy*, $p \approx .07$), whereas found the Naïve UCB robot more annoying (*Annoying*, $p \approx .05$). They also indicated the Naïve UCB insisted more on the suboptimal burgers (*Insisting*, $p < .005$). Finally, the users think they could have more satisfied customers if they were stacking burgers alone with a higher confidence when partnered with the Naïve UCB robot (*Alone*, $p < .05$). These results strongly support **H2**. We further analyze and discuss how different populations of human users perform differently in Appendix ??.

7 Conclusion

Summary. We studied multi-agent decentralized MAB, where the reward obtained by the team depends on all agents’ actions. We showed naïve extensions of optimal single-agent MAB algorithms – where each agent disregarded others’ actions – fail when rewards are coupled. We proposed a simple yet powerful algorithm for partners to model and coordinate with the partners who have higher observability over the task. Our algorithm only relies on the observation of partner’s actions and accomplishes the coordination without explicit communication. We analytically showed it achieves logarithmic regret and tested our hypotheses through simulations and experiments.

Limitations and Future Work. The decentralized MAB is a useful abstraction for many real-world coordination tasks, and we are excited that our algorithm yet simple demonstrates significant improvements to enable seamless coordination. However, many applications require more complex formulations such as Markov Decision Processes. In the future, we plan to extend the intuitions gained by our algorithm and analysis to some of these more complex settings.

Another interesting direction is pairing the partner-aware strategy with algorithms other than UCB, e.g., Thompson sampling. Our preliminary results indicate it still gives significant improvements over the naïve counterparts.

Acknowledgments

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